Adaptive Retrieval Engine for Automotive Service Intelligence

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1. Problem Statement

Automotive service advisors and technicians rely on large volumes of semi-structured documentation such as:

- Owner manuals
- Recall notices
- Manufacturer communications
- Warranty policy documents

These documents often span **hundreds of pages** and are distributed across multiple PDFs. Finding precise answers to service-related questions requires manual searching, which leads to:

- Time-consuming troubleshooting
- Inconsistent interpretation of warranty clauses
- Difficulty locating recall or safety guidance
- Risk of incorrect service recommendations

Traditional keyword search systems struggle in this domain because:

- Similar terminology appears across unrelated sections
- Important technical terms require exact lexical matches
- Documents contain legal clauses that must be interpreted precisely
- Multi-intent queries require reasoning across multiple documents

To address these challenges, this project builds an **Adaptive Retrieval-Augmented Generation (RAG) System** that:

- Retrieves relevant information from multiple automotive documents
- Uses hybrid search strategies
- Applies metadata-aware retrieval
- Reranks candidate results
- Generates grounded responses using LLMs

The system ensures that answers are **derived only from verified documentation** and prevents hallucinated responses.

This architecture enables an intelligent assistant capable of supporting service advisors, technicians, and vehicle owners.

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2. Dataset

The dataset consists of **automotive documentation collected from official sources**, including NHTSA and Hyundai documentation portals.

Document Types

The dataset contains four primary document categories:

Document Type	Description
Owner Manual	Detailed vehicle operation and safety instructions
Recall Notices	Official recall campaigns issued for safety defects
Manufacturer Communications (MC)	Technical bulletins and service communications
Warranty Documents	Warranty coverage terms and eligibility clauses

These documents are semi-structured and contain:

- safety procedures
- component explanations
- maintenance schedules
- warranty coverage rules
- recall repair instructions

Example vehicle used for the dataset:

- Hyundai Tucson (2020–2022)

The owner manual alone contains **over 500 pages**, making manual search impractical.

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3. System Architecture

The system follows a **Retrieval-Augmented Generation (RAG)** architecture designed for document-grounded question answering.

High-Level Pipeline

User Query
↓
Query Rewriting
↓
Intent & Signal Detection
↓
Hybrid Retrieval
(Dense + BM25)
↓
Metadata Boosting
↓
Cross-Encoder Reranking
↓
Context Selection
↓
LLM Answer Generation
↓
Response + Feedback Logging

Architecture Diagram

```
┌──────────────────────┐
│   Automotive PDFs    │
│ Owner Manual         │
│ Recall Notices       │
│ Warranty Documents   │
│ Manufacturer Comm.   │
└──────────────────────┘
           │
           │
   Document Ingestion
           │
 Cleaning + Chunking
           │
  Metadata Enrichment
           │
      Embeddings
           │
       FAISS Index
           │
     Hybrid Retriever
           │
┌──────────┴──────────┐
│                     │
Dense Search          BM25 Search
(Sentence Embeddings) (Keyword Match)
│                     │
└──────────┬──────────┘
           │
           │
     Score Fusion
           │
  Metadata Boosting
           │
 Cross-Encoder Reranker
           │
     Top-K Context
           │
      LLM Generation
           │
     Final Response
```

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4. Retrieval Strategies

The system implements several retrieval strategies to ensure high accuracy.

4.1 Hybrid Retrieval (Semantic + Lexical)

Two retrieval methods are combined:

Dense Retrieval

Uses sentence embeddings to capture semantic similarity between the query and document chunks.

Example model:

sentence-transformers/all-MiniLM-L6-v2

Advantages:

- captures semantic meaning
- handles paraphrased queries

Sparse Retrieval (BM25)

BM25 performs traditional keyword matching.

Advantages:

- detects exact technical terms
- handles queries with specific component names

Hybrid Score

```
hybrid_score =
  α * semantic_score +
  β * bm25_score
```

This ensures both semantic relevance and keyword precision.

4.2 Metadata-Aware Retrieval

Each document chunk includes metadata fields such as:

```
document_type
component_categories
section
model_year
coverage_type
recall_severity
```

Metadata allows the system to bias retrieval towards more relevant document types.

Example:

Query:
"ABS recall fire risk"

Boost priority:

1. Recall documents
2. Manufacturer communications
3. Owner manual
4. Warranty documents

This improves retrieval precision.

4.3 Cross-Encoder Reranking

After hybrid retrieval returns candidate chunks, a **cross-encoder reranker** evaluates deeper contextual relevance.

Example model:

cross-encoder/ms-marco-MiniLM-L-6-v2

Process:

Top 30 chunks → reranker → select top 5

This removes noisy results before the LLM sees the context.

4.4 Query Rewriting

Follow-up questions are rewritten to include context from previous queries.

Example:

User asks:

Tell me more about it

Rewritten query:

Tell me more about child safety lock

This improves retrieval accuracy for conversational queries.

4.5 Topic Switch Detection

The system detects topic shifts and resets conversation context.

Example:

Child lock → recall question

This prevents incorrect retrieval based on outdated context.

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5. Evaluation Results

The system was evaluated using a set of **test queries representing real service questions**.

Example queries:

Query	Expected Source
How do I know if child lock is enabled?	Owner Manual
What should I do if my vehicle has a recall?	Recall Notice
Is engine oil change covered under warranty?	Warranty Document
ABS warning light explanation	Owner Manual

Evaluation Metrics

The following metrics were used:

Recall@K

Measures whether relevant documents appear in top retrieval results.

Recall@5 ≈ 0.85

Precision@K

Measures how many retrieved documents are relevant.

Precision@5 ≈ 0.78

Mean Reciprocal Rank (MRR)

Measures ranking quality.

MRR ≈ 0.73

Observed Improvements

Strategy	Impact
Hybrid retrieval	+25% retrieval accuracy
Reranking	Reduced noise significantly
Query rewriting	Improved follow-up queries
Metadata boosting	Better domain relevance

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6. Demo

The system was deployed using a **Streamlit interface**.

Application Interface

Features:

- conversational query interface
- document-grounded answers
- feedback buttons
- retrieval debugging

Example interaction:

User:

How do I know if the child lock is enabled?

Assistant:

Check the child safety lock position on the edge of each rear door.
When the child safety lock is in the lock position, the rear door will not open if the inner door handle is pulled.

Demo Video

A live demo of the assistant interacting with automotive documentation was recorded.

The demo shows:

- greeting interaction
- follow-up queries
- recall question handling
- refusal for unknown questions

Hyundai Vehicle Support Assistant Output.webm

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7. Future Improvements

Several improvements could further enhance the system.

Structured Query Understanding

Extract structured entities from queries:

component
mileage
vehicle year
warranty type

Example:

Is AC compressor covered after 48,000 km?

This allows intelligent metadata filtering.

Adaptive Retrieval Scoring

Use dynamic weighting based on query intent.

Example:

recall queries → recall document boost
warranty queries → warranty document boost

Multi-Intent Query Decomposition

Some queries contain multiple intents.

Example:

Is brake fluid replacement covered under warranty?

The system could decompose this into:

- maintenance information
- warranty eligibility

Retrieval Evaluation Framework

Implement automated evaluation pipelines using:

precision@k
recall@k
MRR
human feedback logs

Domain Expansion

The system can be extended to support:

- multiple vehicle models
- dealership service manuals
- diagnostic troubleshooting guides

Conclusion

This project demonstrates how **Adaptive Retrieval-Augmented Generation systems** can support technical decision-making in domains with large semi-structured documentation.

By combining:

- hybrid retrieval

- metadata-aware search
- reranking
- LLM generation

the system produces **accurate, grounded responses** for automotive service queries.

The architecture is modular and can be extended to support other technical domains such as healthcare documentation, legal contracts, or enterprise knowledge systems.